Suho Yu

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Research Interest

UAM • eVTOL system integration • High-voltage battery architecture • Autonomous aerial robotics
Embedded control • Fault-tolerant design • Certification & safety • HIL testing • Edge AI using Jetson • MDO

Education

Inha University — Mar 2018~Aug 2023
BS in Mechatronics Engineering — Incheon, KR
Courses Control Systems, Mechatronics Design, UAV System Engineering
Embedded Systems, Signal Processing, Dynamic Systems

Research Experience

eVTOL Integration	Intelligent On-board Avionics SW for Emergency Detection & Response *VSPACE, System Lead – Fault detection avionic software integration*	Apr 2024~Present
	Future Air Mobility UAM Prototype – Development and Flight *VSPACE, Project Lead – System architecture & experimental validation*	Jan 2022~Dec 2024
	Urban Air Mobility (UAM) Type Certification Research *VSPACE, Propulsion Integration Lead - Propulsion architecture design*	Apr 2024~Present
Battery System & Integration	Korea's First Manned All-Electric GA Aircraft Flight – Full Technical Research *VSPACE, Battery System Lead – High-voltage pack design and integration*	Apr 2020~Dec 2023
	Development of 250 Wh/kg-class battery system for UAM applications *VSPACE, Principal Investigator – High-voltage pack design and integration*	Apr 2020~Dec 2023
	High-discharge Battery System for 200 kg Hydrogen-Electric Cargo UAV *VSPACE, Battery System Lead – High-voltage pack design and integration*	Apr 2021~Sep 2023
	Development of Battery System for 40kg-class Heavy-Lift Multirotor Drone *VSPACE, Battery System Lead – High-voltage pack design and integration*	Apr 2021~Sep 2023
	Development of UAV System for Maritime Safety and Ecosystem Monitoring *VSPACE, Battery System Lead – Electric propulsion system integration*	Apr 2021~Sep 2024

Publications & Conference Contributions

Journal Articles	Design, Fabrication, and In-Flight Demonstration of a 24S NCM Battery System for an eVTOL Aircraft *Batteries, 12(9), 317; DOI: 10.3390/batteries12090317 — **First Author***	Aug 2026
	SLM SUS316L Impeller-Based Electric Waterjet: Design and Thrust Verification *Design & Manufacturing, 20(3), 1–13; DOI: 10.22847/ksdme.20.3.202609.001 — **First Author***	Sep 2026
	Modification and Development of Manned Electric Propulsion Lightweight Airplane *Journal of Aerospace System Engineering, 17(3), 99–107; DOI: 10.20910/JASE.2023.17.3.99 — **Co-author***	Jun 2023
	Bench-to-Flight Validation of a Passively Air-Cooled 90 Ah Battery Module for Cargo eVTOL: Power Margin and Post-Landing Heat-Soak Characteristics *IEEE Transactions on Transportation Electrification (Major Revision invited) — **First Author***	Oct 2026 (Major Revision)

	Polarization-Induced SOC Mis-estimation in a Manned Electric Aircraft Battery: Cell Characterization, First-Flight Evaluation, and 2RC-EKF Correction *Energy Storage (Wiley; submitted Aug 2026) — **First Author***	Aug 2026 (Submitted)
Conference Papers & Presentations	Flight-Test-Based Evaluation of Voltage-Sag-Induced Hover Power Margin Reduction in a 210 kg-Class Multicopter eVTOL *AIAA SciTech Forum 2027 — Accepted abstract; oral presentation scheduled — **First Author***	Jan 2027
	Development and Flight Validation of a 250 Wh/kg-Class, 8C-Rate Lithium-Ion Battery Module for Urban Air Mobility Applications *APISAT 2026, Sapporo, Japan — Accepted; **First Author***	Oct 2026
	Experimental Investigation of Thermal Runaway Propagation in Lithium-ion Battery Modules for Electric Propulsion Systems: A Comparative Study of Two Packing Configurations *AJSAE 2026, Jeju, Republic of Korea— **First Author***	Sep 2026
	Flight Log based Analysis of Test Environment Effects on Navigation and Control Behavior in eVTOL Tethered Hover Tests *IAA SpaceAI 2026, Jeju, Republic of Korea— **First Author***	Aug 2026
	Voltage Sag Mitigation through Parallel Battery Configuration and Its Effect on Takeoff Power Margin in GA-Class eCTOL Electric Aircraft *SASE 2026 Spring Conference, Busan— **First & Corresponding Author***	May 2026
	Staged Tethered Flight Testing and log-based Performance Validation of a Manned Multirotor eVTOL System *SASE 2026 Spring Conference, Busan — **First Author***	May 2026
	Design of a Power System for Handling Regenerative Braking Back EMF in a 1000 kW Class Electric Test Facility *SASE 2025 Fall Conference, Jeju — **First Author***	Sep 2025
	Integrated System Design and Flight Demonstration for Ensuring Flight Continuity in an X8 Electric Propulsion Aircraft under Single Propeller Failure *SASE 2025 Spring Conference, Yeosu — **First Author***	May 2025
	A Methodology for Specifying Electric Propulsion Battery Requirements Based on K-UAM Operational Scenarios *SASE 2025 Spring Conference, Yeosu — **Co-author***	May 2025
	Effects of Internal Resistance Deviation on Thermal Performance of High Energy Density Lithium-ion Battery Module *KASA 2023 Spring Conference, Jeju — Conference abstract — **First Author***	May 2023
	Thermal Analysis of the high C-rate Battery Module in an Urban Air Mobility *KSAE 2022 Autumn Conference, p. 2418, 22AKSAE_P043 — Conference abstract — **Co-author***	Nov 2022

Patents

eVTOL Integration	Electronic Device for Providing Emergency Situation Detection and Automatic Response Control and Operating Method Thereof *KR App. 10-2026-0192178; U.S. application filed*	Oct 2026
	Electronic Device for Providing Uninterrupted Energy Transition and Emergency Landing and Operating Method Thereof *KR App. 10-2026-0192179; U.S. application filed*	Oct 2026
	Electronic Device for Providing Automatic Flight Control in Event of Thruster Failure and Operating Method Thereof *KR App. 10-2026-0192177*	Oct 2026
	Device and Method for Detecting Anomalies Using Multivariate Sensor Data by Flight Phase of a Drone *KR App. 10-2025-0160057*	Oct 2025
	Vibration-Absorbing Mounting Structure for Flight Control Computer *KR App. 10-2025-0164972*	Nov 2025
	Apparatus and Method for Active Output Control Responding to Abnormal State of a Flying Object Power System *KR App. 10-2025-0164973*	Nov 2025
	Method and Apparatus for Controlling Flight Vehicle Landing Based on Multi-Sensor Reliability *KR App. 10-2025-0164974*	Nov 2025
	Flight Control Computer Housing With Integrated Heat Dissipation and Electromagnetic Shielding Functions *KR App. 10-2025-0175886*	Nov 2025
	Flight Control Computer Having Redundant CAN Communication Structure and Operating Method Thereof *KR App. 10-2025-0177688*	Nov 2025
	Method and Apparatus for Automatic Power Source Switchover With Multiple Inputs and Flight Control Computer Including the Same *KR App. 10-2025-0177689*	Nov 2025
	Frame (Urban Air Mobility Aircraft) *U.S. Design Patent D1,081,822 S (granted Jul 1, 2025); App. 29/860,942*	Jul 2025
Battery System & Integration	Electronic Device for Automated Compliance Verification of an Aviation Battery Against Certification Requirements and Inverse Computation of Design Specifications for Non-Compliant Items and Operating Method *KR Patent Application Filed (Oct 2026) — Application No. to be confirmed*	Oct 2026
	Battery Module for Electric Propulsion Aircraft and Battery Pack Including the Same *KR App. 10-2025-0152733; U.S. patent application filed (Oct 2025)*	Oct 2025
	High-Power Battery Package for eVTOL and Its Management Method *KR App. 10-2024-0200697*	Dec 2024
	Air-Cooled High-Density Battery Pack *KR App. 10-2024-0171518*	Nov 2024
	Method for an Aircraft to Manage the Battery in Response to Take-Off and Landing *KR App. 10-2024-0004711*	Jan 2024
	Method for an Aircraft to Manage the Battery in Accordance with the Flight Plan That Includes Take-Off and Landing *KR App. 10-2024-0004710*	Jan 2024
	High-Power, High-Density Battery Modules for Future Transportation Devices *KR App. 10-2024-017520*	Dec 2024
	Movable Charger with Improved Charging Speed and Safety *KR App. 10-2023-0163497*	Nov 2023
	Fire Protection Pad and Battery Cell Array Using the Same *KR Patent 10-2952500 (granted Apr 2026); App. 10-2023-0054985*	Apr 2023
	Damper for Reducing Vibration in Two-Wheeled Vehicles *KR Patent 10-2512287 (granted Mar 16, 2023); App. 10-2021-0063107*	Mar 2023

Professional Experience

VSPACE Co., Ltd. Nov 2021~Present
Director of Engineering Incheon, KR

CEAS Co, Ltd. Jul 2018~Jan 2021
Design Engineer Ansan, KR

ROBO Co., Ltd. Mar 2017~Jan 2021
Production Manager Seoul, KR

SEC Co., Ltd. Jan 2016~Feb 2017
Design Engineer Suwon, KR

Awards & Achievement

CNC Milling, 2026 Korea National Skills Competition – Judge Aug 2026
Korea Skills Promotion Committee

Aerospace Technology-Based Start-Up Academy, 2022 – Best Award Oct 2022
Korea Aerospace Research Institute

CNC Milling, 2015 Korea National Skills Competition – Gold Medal Apr 2015
Korea Skills Promotion Committee

CNC Milling, 2023 Korea National Skills Competition – Judge Oct 2023
Korea Skills Promotion Committee

CNC Milling – Player Coaching (192hr) Dec 2023~Feb 2024
Korea Aviation High School

Skills

Tools: SolidWorks, CATIA V5, AutoCAD, Master CAM, STAR-CCM+, Altium Designer STM32CubeIDE
Programming: C, C++, Embedded C, Python
Certification: Engineer – General Machinery, Industrial Safety, Electricity, 3D Printer Development, Computer Aided Manufacturing
Craftsman – Computer Aided Milling, Computer Aided Lathe, Fork Lift Operator, Hydro-pneumatic, 3D Printer Operation
Languages: Korean (Native), English (Advanced), Japanese (Advanced)